AARAV KHANAL

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SUMMARY

B.E. Artificial Intelligence and Machine Learning student graduating in 2028, seeking Software Engineering or AI Engineering internships. Builds AI systems spanning LLM applications, retrieval-augmented generation, developer tools, and computer vision.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL
AI/ML: PyTorch, LLM applications, RAG, multi-agent systems, NLP, computer vision
Frameworks: FastAPI, LangGraph, React, Next.js, scikit-learn
Tools: Git, Docker, Linux, PostgreSQL, SQLite, REST APIs

PROJECTS

AARAV-LLM - Decoder-Only Language Model

Technologies: Python, PyTorch, FastAPI | GitHub: github.com/Aaravkhanal/MY-LLM

- Trained four decoder-only models up to 48.1M parameters on an 8 GB MacBook Air and compared outputs on fixed evaluation prompts.
- Implemented byte-level BPE, RoPE, GQA, resumable pretraining, SFT/LoRA, and web serving; validated the stack with 564 automated tests and push-triggered CI.

ProjectMind AI - Multi-Agent Code Review Platform

Technologies: Python, FastAPI, LangGraph, SQLite | GitHub: github.com/Aaravkhanal/ProjectMind-AI | Published: VS Code Marketplace listing

- Built parallel architecture, security, and quality review agents that synthesize findings into one review with persistent repository context.
- Implemented sub-200 ms environment discovery, encrypted local keys, and budget-aware routing across 15+ LLM providers.

VIBE-SHIELD - Autonomous Security & Quality Scanner

Technologies: Node.js, Playwright, Multi-Agent Systems | GitHub: github.com/Aaravkhanal/VIBE-SHIELD

- Built a six-agent CLI that scans web apps for quality, security, business-logic, API/auth, and prompt-injection risks, producing a consolidated assessment.
- Correlated issues into attack-chain reports and added SARIF/JSON/Markdown/HTML exports plus GitHub Actions for CI-ready reporting.

FrameFlow AI - Design-to-Code Consensus Engine

Technologies: React, FastAPI, Supabase | GitHub: github.com/Aaravkhanal/Frameflow-ai

- Generated frontend code from screenshots, Figma designs, URLs, and screen recordings using OpenAI, Claude, and Gemini consensus.
- Added asset extraction, responsive previews, and multi-file export; documented 228 backend tests for the service workflows.

CODEXA - Terminal-Native AI Coding Agent

Technologies: TypeScript, Bun, MCP, Multi-Agent Systems | GitHub: github.com/Aaravkhanal/CODEXA | Published: npmjs.com/@aaravkhanal/codexa

- Implemented PLAN/BUILD modes and a six-stage Explorer-to-Reviewer workflow so users can inspect planned edits before execution.
- Published an installable TypeScript/Bun CLI with MCP, code search, project memory, and local/cloud model support for terminal-based development.

Nexora AI - Agentic Company Intelligence & RAG

Technologies: Python, FastAPI, ChromaDB, Next.js | GitHub: github.com/Aaravkhanal/NEXORA-AI

- Built parallel agents for company, financial, product, competitor, and SWOT research and streamed progress to an interactive dashboard.
- Indexed findings in company-specific ChromaDB stores to ground follow-up answers in retrieved company sources.

RESEARCH

Canary Memorization Study - Small Language Models

Technologies: Python, PyTorch, Experimental Design | GitHub: github.com/Aaravkhanal/canary-memorization-study

- Designed a reproducible 24-cell experiment across 3 seeds, 4 architectures, and 14M/48M parameter scales to compare memorization exposure.
- Analyzed 960 observations and measured a 1.223-bit exposure increase per tenfold rise in canary frequency (95% CI: 0.691-1.755).

EDUCATION

B.M.S. College of Engineering, Bengaluru | Expected Graduation: 2028
B.E. in Artificial Intelligence and Machine Learning | CGPA: 8.0/10
Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Machine Learning, Deep Learning, Computer Vision, NLP, Software Engineering